Error Analysis

Introduction

The purpose of this error analysis is to better understand where the emotion classification model performs poorly and why. By analyzing the confusion matrix, frequent misclassification patterns, and specific examples of incorrect predictions, we can identify which emotions are harder to classify and the possible reasons for these difficulties. This will help us improve future model performance by highlighting which areas need attention.

Dataset Imbalance and Distribution

One key factor affecting the model's performance is the class imbalance in the dataset. The distribution of emotions is not uniform, which impacts how well the model learns to predict less frequent classes. For example, the *neutral* and *happiness* classes dominate the dataset, with significantly higher sample counts compared to underrepresented emotions like *disgust*, *fear*, and *sadness*. Here is the distribution for the test set:

Emotion	**Samples**
Neutral	440
Happiness	329
Surprise	155
Anger	39
Sadness	34
Disgust	17
Fear	16

The imbalance in the dataset leads to biased predictions, where the model is more likely to predict *neutral* or *happiness* and less likely to correctly identify the underrepresented classes.

Confusion Matrix Overview

The confusion matrix reveals several important insights about the model's performance across different emotion classes. Firstly, the model performs well with the *happiness* class, correctly predicting it 230 times, and *neutral*, with 343 correct predictions. However, there are notable patterns of misclassification. *Neutral* is frequently misclassified as *happiness* (40 misclassifications) or *surprise* (32 misclassifications). On the other hand, *happiness* is often misclassified as *neutral*, *surprise*, and occasionally *disgust*. The model faces significant challenges with underrepresented classes like *disgust* and *fear*, with these classes being predicted correctly only 5 and 8 times, respectively. Additionally, there are symmetric confusions, particularly between *neutral*, *happiness*, and *surprise*, where the model struggles to differentiate these emotions, especially when the context is subtle.

Key observations from the confusion matrix show that *neutral* is often confused with *happiness* and *surprise*, while *happiness* is frequently misclassified as *neutral*, *surprise*, and even *disgust*. Additionally, underrepresented classes such as *disgust* and *fear* are seldom predicted correctly. These misclassification patterns highlight the difficulties the model faces, particularly with emotions that are less frequent in the training dataset. To improve the model's performance on these underrepresented classes, a more balanced dataset or techniques like oversampling may be needed.

Example of misclassifications:

- **True label: Neutral**
  - Predicted as: Surprise (e.g., “gata!”)
  - Predicted as: Disgust (e.g., “do not hold your hands, that your hands help you hold on to the bottom”)
- **True label: Happiness**
  - Predicted as: Disgust (e.g., “the diandel and larisa resist skating in the sand”)
  - Predicted as: Neutral (e.g., “okay, okay, okay!”)
- **True label: Sadness**
  - Predicted as: Surprise (e.g., “but you were already in a bottom, you were just”)
  - Predicted as: Disgusted (e.g., “no, we are not in advantage”)
- **True label: Fear**
  - Predicted as: Happiness (e.g., “to me, vladu! to me!”)
  - Predicted as: Surprise (e.g., “head below, ina! head below, ina!”)
- **True label: Anger**
  - Predicted as: Fear (e.g., “but to attack”)
  - Predicted as: Neutral (e.g., “no more! neither marinescu!”)
- **True label: Surprise**
  - Predicted as: Fear (e.g., “try here is out of mind”)
  - Predicted as: Happiness (e.g., “we win!”)
- **True label: Disgust**
  - Predicted as: Anger (e.g., “they did nothing!”)
  - Predicted as: Fear (e.g., “i said two, three things, punch in my mouth now”)

The model demonstrates more accurate results compared to the generated test set, although this is open to subjectivity. A notable observation is that the "neutral" class is frequently confused with other emotions, which could be due to the quality of the test set itself. These misclassifications suggest that context and word choice play a crucial role in the model's ability to correctly predict emotions. The model tends to mix up similar emotional tones, such as the confusion between "happiness" and "neutral," or "surprise" and "fear." This highlights the need for further refinement in distinguishing these emotions, potentially through more advanced models or additional feature engineering.

Other important observations

Most Frequent Misclassifications

The top-10 misclassifications highlight dominant error patterns, such as "happiness" → "neutral" (53) and "neutral" → "happiness" (40), which suggest a semantic or linguistic overlap between these emotions. Similarly, "surprise" → "neutral" (24) indicates that surprise may not always manifest strongly in text, which can contribute to misclassification. The model seems to default to dominant classes, particularly when it is unsure, often categorizing emotions that share overlapping characteristics. This shows that the model might struggle to differentiate between closely related emotions in subtle contexts.

Impact of Text Length and Ambiguity

The analysis of text length reveals a significant statistical difference in the average length of misclassified versus correctly classified texts. Misclassified texts have an average length of 7.05 tokens, while correctly classified texts average 5.70 tokens. This suggests that longer texts may introduce more ambiguity, potentially due to mixed sentiments, sarcasm, or shifts in tone. The length and complexity of sentences appear to affect the model's ability to predict emotions accurately, possibly causing it to misinterpret ambiguous emotional cues.

Qualitative Analysis by Class

A more detailed qualitative analysis of misclassifications by emotion provides additional insights into the model's performance, especially when considering the origin of the test set. For example, in the case of "happiness" → "neutral," consider a sentence like "and here, he pulls," which might be predicted as neutral by the model. The implied happiness in this case is subtle, context-dependent, and not explicitly stated, which makes it challenging for both the model and humans to identify the correct emotion. However, it’s worth noting that the test set itself, created using LLMs and a script from a YouTube video, may have inconsistencies in labeling subtle or context-dependent emotions. The model, in this case, might actually perform better by correctly identifying the implied emotion as neutral, as human interpretation of such cues can vary widely.

Similarly, a "surprise" → "neutral" misclassification, such as “I can’t believe it’s happening,” might seem like surprise to a human reader but could be predicted as neutral by the model. In this case, the LLM-generated test set correctly labels the text as "surprised," capturing the implied emotion, whereas the model fails to pick up on the subtle emotional tone. The absence of strong, overt emotional cues in the text likely led the model to classify the statement as neutral.

In instances where the model confuses similar emotions—such as "happiness" → "neutral" or "surprise" → "fear"—the test set might be less reliable in assigning clear labels, especially since emotions like surprise or happiness can be expressed with subtlety. For example, in a sentence like "I was just so overwhelmed, I couldn’t stop laughing,” the implied surprise and happiness may overlap, and the test set might not have captured this ambiguity correctly. Here, the model might provide a better interpretation of the subtle emotional mix. This could be due to the limitations of the LLM-generated test set, which may lack the ability to account for metaphorical or context-driven emotional expression.

These examples suggest that while misclassifications still occur, the model may sometimes offer more accurate predictions in cases where the test set fails to adequately handle the complexity and nuance of emotion in text. This underscores the importance of refining both the test set and the model to better capture implied emotions, ambiguity, and subtlety in text, which will ultimately improve performance.

Challenges in Emotion Detection from Text

General challenges in emotion detection include the contextual and subjective nature of emotions, where lack of punctuation, emojis, or tone of voice can hinder accurate interpretation. Emotions like "disgust," "fear," and "surprise" are often reliant on physical cues that text alone cannot convey. Additionally, polysemous expressions or domain-specific phrases (e.g., "punch in my mouth" ≠ literal) complicate the labeling process, as the model may struggle to differentiate between literal and figurative language.

Suggestions for Improvement

To address these challenges, several improvements can be made. Contrastive learning could help the model better differentiate between close emotions, such as "happiness" vs "neutral." Multimodal features, such as combining text with audio or visual data, could further enhance the model’s performance by providing additional context for emotional cues.

Based on the classification report, we can assess the model's performance by looking at several key metrics like precision, recall, F1-score, and support for each class. Precision measures how many of the predicted positive instances are actually correct. In this report, the classes "happiness" and "neutral" show high precision (0.77 and 0.79, respectively), indicating the model is generally good at identifying instances of these emotions. However, precision for other classes, such as "disgust" (0.15) and "fear" (0.24), is much lower, indicating a higher rate of false positives in these categories.

Recall indicates how many of the actual positive instances are correctly identified. The "neutral" class stands out with a recall of 0.78, which suggests the model is good at capturing most of the instances for this class. The recall for "disgust" (0.29) and "fear" (0.50) is lower, which suggests the model may be missing many instances of these classes.

F1-Score is the harmonic mean of precision and recall, balancing both metrics. "Neutral" again performs well with an F1-score of 0.78. "Happiness" also performs well with a score of 0.73. However, "disgust," "sadness," and "fear" have lower F1-scores, indicating that the model is struggling to balance both precision and recall in these cases.

The overall accuracy of the model is 0.69, which shows the model is correctly classifying about 69% of the instances across all classes, and the F1 score is 0.70 (weighted).

For this task, the F1 score is the most important metric, as it balances both precision and recall. In tasks like emotion detection, it is important to both correctly identify the positive instances (high recall) and avoid misclassifying them as something else (high precision). Focusing solely on accuracy may not give a full picture, as it can be skewed by the dominant classes. In this case, "happiness" and "neutral" have relatively high F1-scores, indicating that the model performs better in these categories, while other classes, such as "disgust" and "fear," suffer from poor recall and precision, lowering the overall F1-score for those categories. Therefore, improving the model's F1 score, especially for underrepresented classes, would be the most beneficial for this task.

Conclusion

In conclusion, the most frequent errors stem from misclassifications between semantically close emotions, as well as dataset imbalance and ambiguity in text. The model's performance highlights the importance of refining emotion detection systems to account for subtle emotional cues, mixed sentiments, and context. Improving the model's ability to distinguish nuanced emotional signals will require more diverse and balanced data, as well as advanced modeling techniques that consider both the content and context of the text.